Unit overview

This unit is an introduction to applied machine learning. Students may have seen Git before.

Learning outcomes

By the end of this unit students will be able to build and evaluate a regression model in Python.

Week 1: Introduction

Students learn Python programming and use pandas and NumPy for data wrangling. Exploratory data analysis is introduced with matplotlib. SQL queries are used to extract data from relational databases.

Week 2: Supervised learning

The unit covers machine learning fundamentals: linear regression, logistic regression and random forests with scikit-learn. Model evaluation uses cross-validation, precision and recall.

Week 3: Deep learning

Neural networks are built with PyTorch. The unit introduces natural language processing, including transformers and large language models, plus the Spark ecosystem for big data. We will spark discussion about ethics.

Week 4: Deployment

Students package models with Docker, deploy on AWS and track experiments with MLflow. R is optional. Ordinary R&D; budgets are out of scope.

Assessment

Assessment 1 is a group project presented in Tableau dashboards. Assessment 2 is a model deployed with Docker.